

1.Data.yaml配置文件代码：

```
path: d:/test
train: images/train
val: images/val
test: images/test
nc: 1
names: ['car']
```

2.test.py程序代码：

```
from ultralytics import YOLO # 加载 YOLOv11n 模型
model = YOLO('yolo11n.pt') # 使用预训练模型
# 开始训练
results = model.train(
    data='data.yaml',      # 数据集配置文件
    epochs=100,             # 训练轮数
    imgsz=640,              # 图像尺寸
    batch=16,               # 批次大小
    device='cpu',           # 设备 (表示 CPU运行)
    name='yolo11n_custom'   # 实验名称, 结果保存在目录
# 可选: 评估模型
metrics = model.val()
print("mAP50-95:", metrics.box.map)
```

3.图片推理picture(hh).py程序代码：

```
from ultralytics import YOLO
import cv2
import os
import numpy as np

def diagnose_model():
    """诊断模型问题"""
    model_path = 'd:/hh/runs/detect/yolo11n_custom/weights/best.pt'
    image_path = 'd:/hh/chepai.jpg'

    # 1. 检查文件是否存在
    print("=== 文件检查 ===")
    if not os.path.exists(model_path):
        print(f"❌ 模型文件不存在: {model_path}")
        return
    else:
        print(f"✅ 模型文件存在: {model_path}")

    if not os.path.exists(image_path):
        print(f"❌ 图像文件不存在: {image_path}")
        return
    else:
        print(f"✅ 图像文件存在: {image_path}")

    # 2. 加载模型并检查信息
    print("\n=== 模型信息 ===")
    try:
        model = YOLO(model_path)
        print(f"✅ 模型加载成功")
        print(f"模型类别: {model.names}")
        print(f"类别数量: {len(model.names)}")
    except Exception as e:
        print(f"❌ 模型加载失败: {e}")
        return

    # 3. 检查图像
    print("\n=== 图像信息 ===")
    img = cv2.imread(image_path)
    if img is None:
        print(f"❌ 无法读取图像")
        return
    else:
        print(f"✅ 图像读取成功")
        print(f"图像尺寸: {img.shape}")

    # 4. 测试不同参数进行推理
    print("\n=== 推理测试 ===")
```

```

# 测试1: 使用极低置信度
print("测试1: 极低置信度 (conf=0.01)")
results = model.predict(source=image_path, conf=0.01, save=False)
boxes = results[0].boxes
if boxes is not None and len(boxes) > 0:
    print(f"✅ 检测到 {len(boxes)} 个目标")
    for i, box in enumerate(boxes):
        cls = int(box.cls[0])
        conf = box.conf[0]
        print(f"  目标 {i + 1}: {model.names[cls]} - 置信度 {conf:.3f}")
else:
    print(f"❌ 未检测到任何目标")

# 测试2: 使用不同图像尺寸
print("\n测试2: 不同图像尺寸")
for imgsz in [320, 640, 1280]:
    results = model.predict(source=image_path, imgsz=imgsz, conf=0.01, save=False)
    boxes = results[0].boxes
    num_detections = len(boxes) if boxes is not None else 0
    print(f"  图像尺寸 {imgsz}: 检测到 {num_detections} 个目标")

# 测试3: 使用保存功能
print("\n测试3: 保存结果")
results = model.predict(source=image_path, conf=0.01, save=True)
print(f"✅ 推理完成, 结果已保存到 runs/detect/predict 文件夹")

# 5. 手动显示结果
print("\n=== 手动显示结果 ===")
if boxes is not None and len(boxes) > 0:
    annotated_img = results[0].plot()
    cv2.imshow('Detection Results', annotated_img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print(f"❌ 没有检测结果可显示")

def check_training_quality():
    """检查训练质量"""
    print("\n=== 训练质量检查 ===")
    model_path = 'd:/hh/runs/detect/yolo11n_custom/weights/best.pt'

    try:
        model = YOLO(model_path)

        # 创建一个简单的测试图像
        test_img = np.ones((640, 640, 3), dtype=np.uint8) * 128
        results = model.predict(source=test_img, conf=0.01, save=False)

```

```

boxes = results[0].boxes
if boxes is not None and len(boxes) > 0:
    print("✅ 模型可以检测目标")
    for box in boxes:
        cls = int(box.cls[0])
        conf = box.conf[0]
        print(f"  检测到: {model.names[cls]} - 置信度: {conf:.3f}")
else:
    print("❌ 模型无法检测任何目标")
    print("可能原因:")
    print("  1. 训练数据不足")
    print("  2. 训练epoch不够")
    print("  3. 模型没有正确学习")
    print("  4. 类别定义有问题")

except Exception as e:
    print(f"检查失败: {e}")

```

```

def alternative_detection_method():
    """备选检测方法"""
    print("\n=== 备选检测方法 ===")
    image_path = 'd:/hh/chepai.jpg'

    img = cv2.imread(image_path)
    if img is None:
        print("无法读取图像")
        return

    # 使用预训练的YOLO模型
    print("尝试使用预训练的YOLOv8模型...")
    pretrained_model = YOLO('yolov8n.pt')

    # 检测车辆
    results = pretrained_model(img, conf=0.3)

    vehicles_detected = 0
    for result in results:
        if result.boxes is not None:
            for box in result.boxes:
                cls = int(box.cls[0])
                class_name = pretrained_model.names[cls]
                if class_name in ['car', 'truck', 'bus']:
                    vehicles_detected += 1
                    print(f"检测到车辆: {class_name}")

    if vehicles_detected > 0:
        print(f"✅ 预训练模型检测到 {vehicles_detected} 个车辆")

```

```

        annotated_img = results[0].plot()
        cv2.imshow('Pretrained Model Detection', annotated_img)
        cv2.waitKey(0)
        cv2.destroyAllWindows()
    else:
        print("❌ 预训练模型也未检测到车辆")
        print("可能图像中确实没有车牌/车辆，或者图像质量有问题")

if __name__ == "__main__":
    print("开始诊断车牌识别问题...")
    diagnose_model()
    check_training_quality()
    alternative_detection_method()

```

4.视频推理video1.py程序代码：

```

import cv2
from ultralytics import YOLO
# 加载型
model = YOLO('d:/hh/runs/detect/yolo11n_custom/weights/best.pt')
# 视频文件
video_path="d:/hh/1.mp4"
# 打开视频
cap = cv2.VideoCapture(video_path)
while cap.isOpened():
    #获取图像
    res, frame = cap.read()
    # 如果读取成功
    if res:
        # 正向推理
        results = model(frame)
    #绘制结果
    annotated_frame = results[0].plot()
    #显示图像
    cv2.imshow(winname="YOL011",mat=annotated_frame)
    # 儻鷁峯小按ESC退出
    if cv2.waitKey(1) == 27:
        break
    else:
        break
# 释放链接
cap.release()
# 销毁所有窗口
cv2.destroyAllWindows()

```